Data Analysis Portfolio

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Professional Background

My name is Gayatri Palkar. I am a highly motivated Data Science professional with over 3 years of experience in the field, currently serving as a Machine Learning Engineer. My expertise lies in utilizing advanced Python libraries such as Pandas, Plotly, and others to deliver impactful data-driven solutions. I possess a solid foundation in data analysis, having successfully completed multiple projects with Trainity, where I honed my skills in extracting insights from complex datasets.

In addition to my proficiency in Python, I have a strong understanding of SQL for database management and querying, along with basic Excel skills for data manipulation and visualization. My experience in the industry has equipped me with the ability to develop and deploy machine learning models, automate data workflows, and collaborate effectively with cross-functional teams to drive decision-making processes.

I am passionate about continuously expanding my knowledge and applying my skills to solve real-world problems through data science and machine learning.

Data Analytics Process

Applying the Data Analytics Process to Real-Life Scenarios

Project Overview:

This project involved creating a case study and presentation that demonstrated how the Data Analytics process can be applied to everyday decision-making. The focus was on illustrating how systematic planning, preparation, and analysis can be used to make informed choices, such as selecting the best shampoo or an online data science course.

Tools & Technologies Used:

- **Microsoft PowerPoint (for creating the presentation)**
- **Microsoft Word or Google Docs (for drafting the case study)**
- **Research and analysis tools (online reviews, product comparison websites)**

Role & Responsibilities:

- **Identified real-life scenarios where the Data Analytics process could be applied.**
- **Broke down each scenario into the steps of the Data Analytics process: Plan, Prepare, Process, Analyze, Share, Act.**
- **Created a detailed case study and corresponding PowerPoint presentation to explain how each step of the Data Analytics process was applied in the scenarios.**
- **Provided examples to illustrate the practical application of data analysis in everyday life.**

Data:

- **Conducted research to gather data on various products (e.g., shampoos, online courses), including customer reviews, ratings, and product specifications.**

- **Analyzed this data to support decision-making in the case study.**

Approach & Methodology:

- **Scenario 1: Choosing the Right Shampoo**
  - **Plan: Identified the key factors to consider when selecting a shampoo, such as hair type and specific hair concerns.**
  - **Prepare: Set a budget and listed any allergies or hair issues to be addressed.**
  - **Process: Gathered and reviewed data on various shampoos, including ingredients, customer reviews, and product ratings.**
  - **Analyze: Narrowed down the options to the top 3-4 products that fit the budget and needs, considering factors like quantity, price, and user feedback.**
  - **Share: Decided on the best option and made a purchase.**
  - **Act: Used the chosen product and evaluated its effectiveness.**
- **Scenario 2: Selecting a Data Science Course**
  - **Plan: Defined learning objectives and goals for taking a data science course.**
  - **Prepare: Established a budget, identified preferred learning methods (e.g., video, hands-on practice), and assessed the current skill level.**
  - **Process: Researched available courses online, considering difficulty level and course content.**
  - **Analyze: Compared courses based on ratings, reviews, content quality, duration, and additional skills taught.**
  - **Share: Selected and enrolled in the most suitable course.**
  - **Act: Began the course and applied the new knowledge to personal projects.**

Key Skills Demonstrated:

- **Application of the Data Analytics process in non-technical, real-life scenarios.**

- **Research and data gathering.**
- **Analytical thinking and decision-making.**
- **Presentation skills**

Results & Insights:

- **Successfully illustrated how the structured approach of Data Analytics can be applied to everyday decisions, making the process more relatable and understandable.**
- **Demonstrated the value of data-driven decision-making in personal and professional contexts.**

Conclusion & Learnings:

- **Gained a deeper understanding of the Data Analytics process by applying it to real-life scenarios.**
- **Improved ability to communicate complex concepts in a simple and relatable manner.**
- **Enhanced skills in creating professional presentations that effectively convey analytical insights.**

Instagram User Analytics

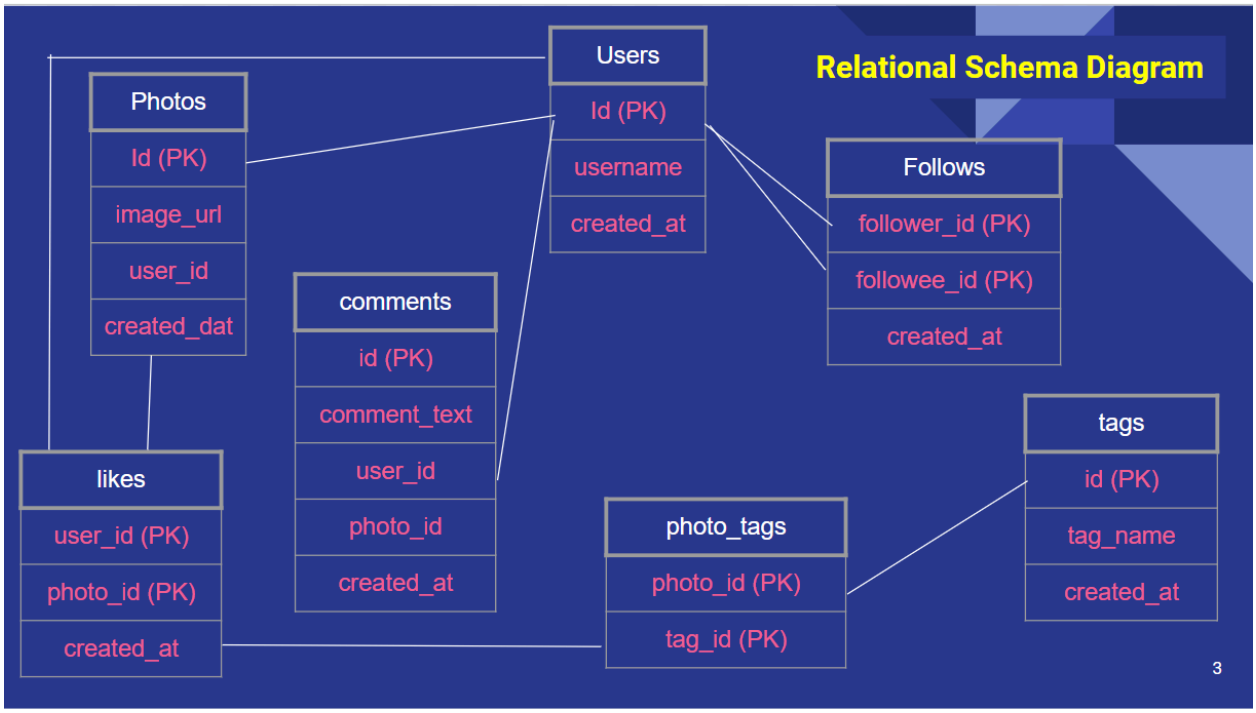
Project Overview: This project involved using SQL and MySQL Workbench to analyze Instagram user data. The goal was to extract meaningful insights that could help the product team at Instagram make informed decisions about the future development of the app. The analysis covered both marketing insights and investor metrics, providing valuable data on user engagement, content popularity, and potential bot activity.

Project Objectives:

- **Conduct user analysis to inform product, marketing, and development teams.**
- **Provide insights on user behavior, engagement, and platform trends.**
- **Utilize SQL queries to extract and analyze data from Instagram's database.**

Tools & Technologies:

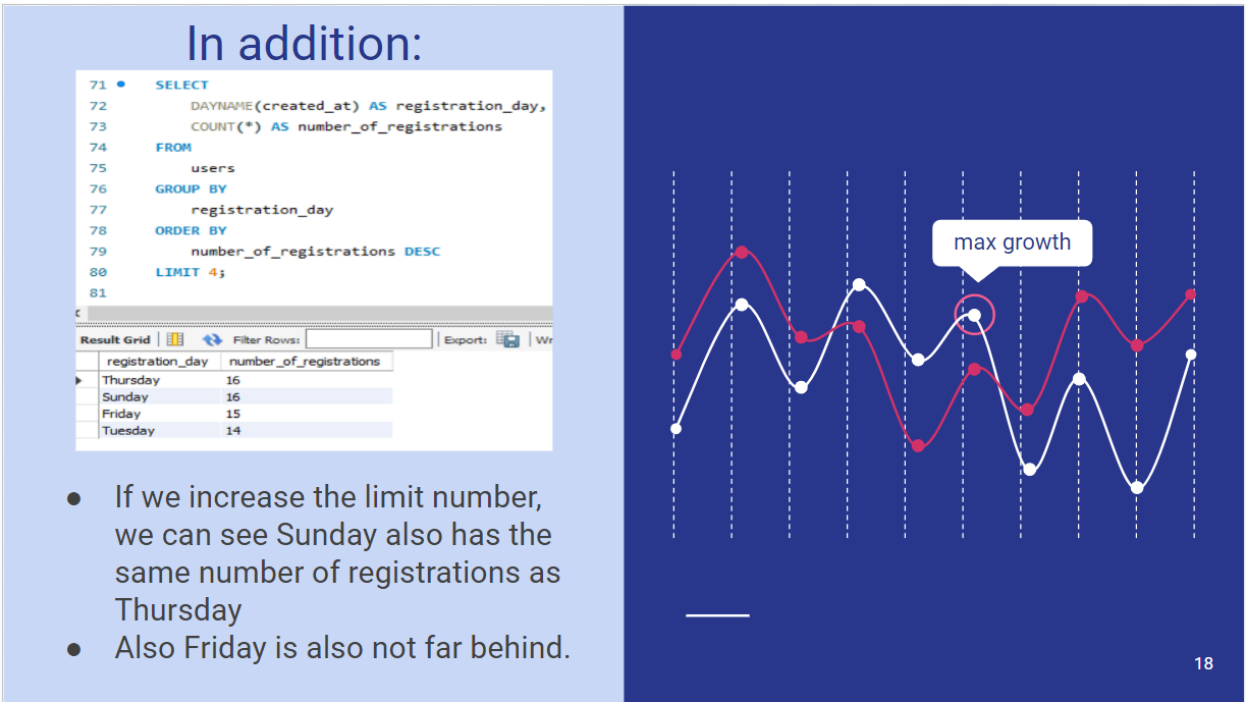
- **SQL**
- **MySQL Workbench**
- **Data Analysis**



Key SQL Tasks:

1. **Loyal User Reward:**
   - **Objective: Identify the five oldest users on Instagram.**
   - **Approach: Selected users from the database, ordered by account creation date, and retrieved the top 5 oldest users.**
   - **Outcome: Provided the marketing team with details of the oldest users for potential rewards.**
2. **Inactive User Engagement - Identify users who have never posted a single photo.**
   - **Approach: Joined the Users and Photos tables using a left join and identified users without any posts.**
   - **Outcome: Offered insights into user inactivity, which can be used to develop engagement strategies.**
3. **Contest Winner Declaration - Determine the user with the most likes on a single photo.**
   - **Approach: Joined Users, Photos, and Likes tables, grouped by photo_id, and identified the photo with the most likes.**

    - **Outcome: Identified the contest winner, helping the team declare the winner.**
4. **Hashtag Research - Suggest the top five most commonly used hashtags.**
    - **Approach: Joined Photo Tags and Tags tables, grouped by tag_name, and ordered by frequency to find the most popular hashtags.**
    - **Outcome: Provided a list of popular hashtags, helping brands and influencers optimize their content for better reach.**
5. **Ad Campaign Launch - Determine the best day of the week to launch ads.**
    - **Approach: Used the Dayname() function to analyze user registrations by day of the week, identifying peak registration days.**
    - **Outcome: Recommended the best day for launching ad campaigns to maximize visibility and engagement.**
6. **User Engagement (Investor Metrics) - Calculate the average number of posts per user.**
    - **Approach: Divided the total number of photos by the total number of users to find the average posts per user.**
    - **Outcome: Provided insights into user engagement, highlighting areas for improvement in content creation.**
7. **Bots & Fake Accounts - Identify potential bot accounts.**
    - **Approach: Found users who liked every single photo on the platform, indicating potential bot activity.**
    - **Outcome: Helped the team identify and address bot activity, ensuring platform integrity.**



Insights & Impact:

- **The analysis helped the Instagram team identify trends, understand user behavior, and address potential issues with user engagement and fake accounts.**
- **The project provided actionable insights for marketing campaigns, user engagement strategies, and platform improvements.**

Conclusion: This project demonstrated the power of SQL in extracting valuable insights from user data, helping to inform business decisions and optimize product development. The findings from this analysis contributed to improving user experience, driving engagement, and maintaining platform authenticity.

Operations and Metric Analytics

Project Overview

I took on the role of a Lead Data Analyst at a company like Microsoft, using advanced SQL skills to analyze various datasets. The primary goal was to investigate metric spikes and provide valuable insights to different departments, improving operational efficiency and understanding user engagement trends.

Case Study 1: Job Data Analysis

Tasks and Insights

1. **Jobs Reviewed Over Time:**
    - **Calculate the number of jobs reviewed per hour for each day in November 2020.**
    - **Approach: I have filtered the data, extracted hourly timestamps, and grouped the data to calculate the number of jobs reviewed.**
    - **Insight: Identified peak review hours, with an average of 1 job reviewed per hour. This could help in resource allocation.**
2. **Throughput Analysis:**
    - **Objective: Calculate the 7-day rolling average of throughput.**
    - **Approach: Calculated daily events and then the 7-day rolling average.**
    - **Insight: The 7-day rolling average provided a clearer trend over time, useful for long-term decision-making.**
3. **Language Share Analysis:**
    - **Objective: Calculate the percentage share of each language in the last 30 days.**
    - **Approach: Filtered the data and calculated language distribution.**

    - Insight: English, Spanish, and French were the top languages, indicating where to focus resources.
4. Duplicate Rows Detection:
    - Objective: Identify duplicate rows in the data.
    - Approach: Grouped data by all columns to find duplicates.
    - Insight: Ensured data quality by identifying and addressing duplicates.

Case Study 2: Investigating Metric Spike

Tasks and Insights

1. Weekly User Engagement:
    - Objective: Measure the activeness of users on a weekly basis.
    - Approach: Extracted weekly data and calculated engagement per user.
    - Insight: Identified most active users, helping to track engagement trends.
2. User Growth Analysis:
    - Objective: Analyze user growth over time.
    - Approach: Tracked new users per month.
    - Insight: Detected growth trends, which could be linked to marketing campaigns or product changes.
3. Weekly Retention Analysis:
    - Objective: Analyze weekly retention of users after signup.
    - Approach: Tracked user retention based on their signup cohort.
    - Insight: Provided insights into user loyalty and areas needing improvement.
4. Weekly Engagement Per Device:
    - Objective: Measure user engagement per device.
    - Approach: Calculated weekly engagement metrics for each device.
    - Insight: MacBook Pro was the most used device, guiding platform-specific optimizations.

5. Email Engagement Analysis:
    - Objective: Analyze user engagement with email services.
    - Approach: Tracked email-related actions per month.
    - Insight: Weekly digests had higher engagement, informing future email campaigns.

Result Grid | Filter Rows: | Export

year	month	action	action_count
2014	8	sent_weekly_digest	16480
2014	7	sent_weekly_digest	15902
2014	6	sent_weekly_digest	13155
2014	5	sent_weekly_digest	11730
2014	8	email_open	5978
2014	7	email_open	5611
2014	6	email_open	4658
2014	5	email_open	4212
2014	7	email_clickthrough	2721
2014	6	email_clickthrough	2274
2014	5	email_clickthrough	2023
2014	8	email_clickthrough	1992
2014	8	sent_reengageme...	1073
2014	7	sent_reengageme...	933
2014	6	sent_reengageme...	889
2014	5	sent_reengageme...	758

- **Insights:**

1. Here we can see that certain email content gets more user engagement. Weekly digests get more clicks out of other emails and reengagement.
2. The results can help to understand user interactions with email communications, such as the number of opens, clicks, bounces, etc.
3. Analyzing these metrics provides insights into the effectiveness of email campaigns and user engagement with email content.

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Conclusion

My analysis provided valuable insights into operational metrics and user engagement. By using SQL effectively, I was able to identify trends, optimize resources, and suggest improvements. This project showcases my ability to handle complex data and derive actionable insights, which is crucial for driving business decisions.

Hiring Process Analysis for a Multinational Company

Overview:

This project aims to analyze the hiring process data of a multinational company. The objective is to draw meaningful insights from the data to help the hiring department improve its strategies and make better hiring decisions. Key areas of focus include understanding gender distribution among hires, analyzing salary trends, examining departmental hiring patterns, and assessing the distribution of positions within the company.

Objectives:

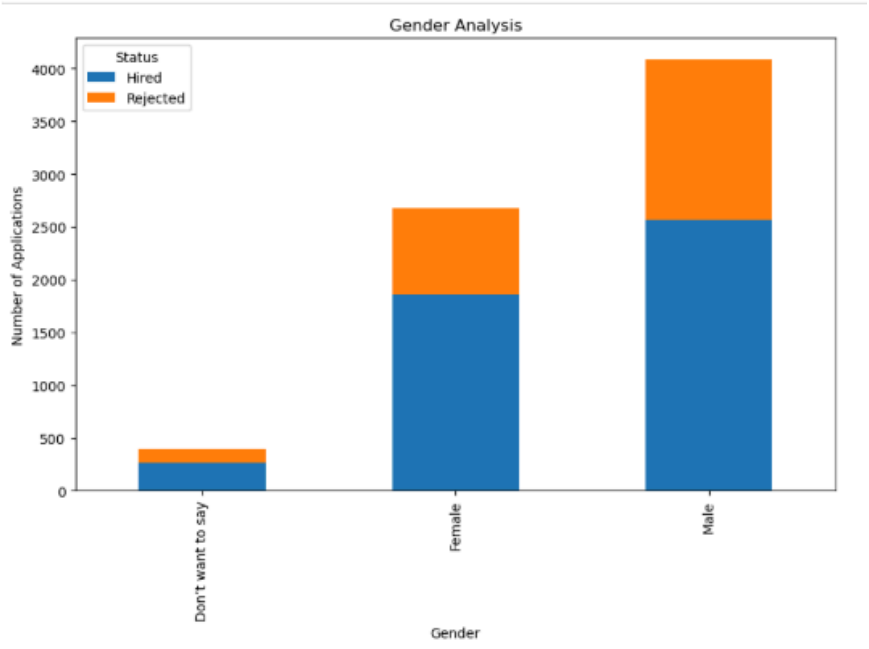
1. **Hiring Analysis: Determine the gender distribution of hires.**
2. **Salary Analysis: Calculate the average salary offered by the company.**
3. **Salary Distribution: Create class intervals for the salaries to understand their distribution.**
4. **Departmental Analysis: Visualize the proportion of people working in different departments.**
5. **Position Tier Analysis: Represent the distribution of positions across different tiers.**

Approach:

1. **Data Preparation:**
    - **Load the dataset into Excel.**
    - **Clean the data by handling missing values and outliers.**
2. **Hiring Analysis:**
    - **Used a pivot table to determine the gender distribution of hires.**
    - **Visualized the gender distribution using a bar chart or pie chart.**
3. **Salary Analysis:**
    - **Calculated the average salary using Excel functions.**

- Visualized the salary distribution using a histogram and a box plot.

4. Salary Distribution:
    - Created class intervals for the salaries.
    - Used the FREQUENCY function to calculate the frequency of salaries within each interval.
    - Visualized the frequency distribution using a histogram.
5. Departmental Analysis:
    - Used a pivot table to analyze the number of hires and rejections across different departments.
    - Visualized the departmental hiring patterns using a bar chart or pie chart.
6. Position Tier Analysis:
    - Used a pivot table to represent the different position tiers within the company.
    - Visualized the distribution of positions across different tiers using a column chart or bar chart.



Tools and Techniques:

- Excel: For data cleaning, analysis, and visualization.
- Pivot Tables: To summarize and analyze data.
- Excel Functions: For calculations (e.g., AVERAGE, FREQUENCY).
- Charts: For visualizing data (e.g., histograms, box plots, bar charts, pie charts).

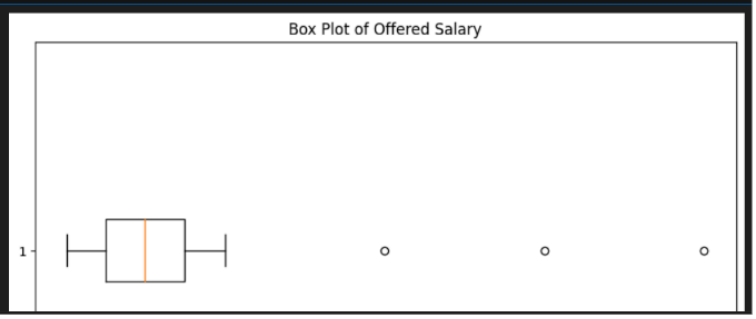
Python code for box plot

```python
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_excel('Statistics_project04_preprocessed1.xlsx')

salary = data['Offered Salary']

plt.figure(figsize=(10, 6))# plotting a box plot to visualize outliers
plt.boxplot(salary.dropna(), vert=False)
plt.title('Box Plot of Offered Salary')
plt.xlabel('Salary')
plt.show()
```



Conclusion:

By analyzing the hiring process data using Excel, the project provides insights into gender distribution, salary trends, departmental hiring patterns, and position distributions. These insights can help the hiring department understand current trends, identify areas for improvement, and make informed decisions to enhance the hiring process.

IMDB Movie Analytics

Overview: The project aims to analyze a dataset of movies from IMDB to gain insights into various aspects of the movie industry. This includes examining the distribution and impact of movie genres, exploring the relationship between movie durations and their IMDB scores, analyzing the influence of languages and directors on movie ratings, and investigating the correlation between movie budgets and their financial success.

Objectives:

1. **Genre Analysis: To understand the distribution of movie genres and their impact on movie ratings.**
2. **Duration Analysis: To analyze the distribution of movie durations and their relationship with IMDB scores.**
3. **Language Analysis: To determine the most common languages used in movies and analyze their impact on IMDB scores.**
4. **Director Analysis: To identify top directors based on their average IMDB score and analyze their contribution to movie success using percentile calculations.**
5. **Budget Analysis: To explore the correlation between movie budgets and gross earnings and identify movies with the highest profit margin.**

Understanding Financial Success

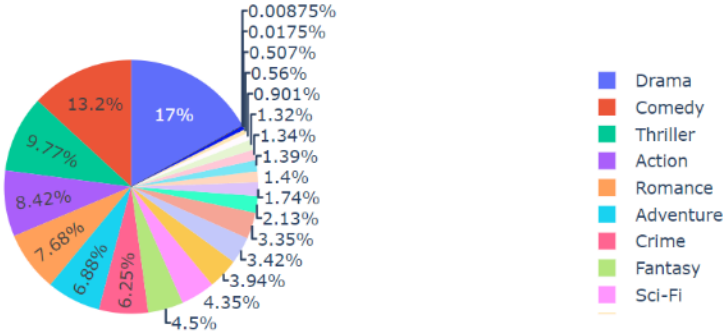
Analyzing the relationship between movie budgets and their financial success is crucial for several reasons:

1. **Investment Decisions:**
    - **Informed Investments: Understanding which types of movies tend to have higher returns on investment (ROI)**

helps producers and investors make more informed decisions.
    - Risk Management: Identifying budget ranges that correlate with higher profits can help mitigate financial risks.
2. Resource Allocation:
    - Optimized Budgeting: Insight into budget effectiveness allows studios to allocate resources more efficiently, ensuring that funds are invested where they are most likely to yield high returns.
    - Strategic Planning: Strategic allocation of funds can be planned based on historical data of successful budget-to-earnings ratios.

2. Key Metrics and Their Importance

1. Correlation Coefficient:
    - Relationship Insight: The correlation coefficient indicates how closely the movie budget and gross earnings are related. A strong positive correlation suggests that higher budgets often result in higher earnings, whereas a weak or negative correlation might indicate other factors at play.
    - Predictive Value: Knowing the correlation helps in predicting potential earnings based on the budget, aiding in financial forecasting and planning.
2. Profit Margin Calculation:
    - Profitability Analysis: Calculating the profit margin for each movie helps in assessing the actual financial success beyond just gross earnings. It reveals how much profit a movie made relative to its cost.
    - Performance Benchmarking: Identifying the movies with the highest profit margins allows studios to benchmark performance and set financial targets for future projects.



3. Strategic Insights

1. **Investment Focus:**
   - **High ROI Movies: By identifying directors or genres associated with high-profit margins, studios can focus on producing similar types of movies that have historically performed well.**
   - **Balanced Portfolios: Studios can create a balanced portfolio of movies, mixing high-budget projects with lower-budget ones to diversify risk and maximize overall profitability.**
2. **Market Trends:**
   - **Trend Analysis: Understanding how budget correlates with earnings over time can help identify market trends, such as increasing costs for high returns or the success of low-budget independent films.**
   - **Audience Preferences: Insights into which budget ranges yield the highest returns can also reflect audience preferences and spending habits, guiding marketing and distribution strategies.**
3. **Competitive Advantage:**
   - **Data-Driven Decisions: Studios that leverage such detailed financial analyses can gain a competitive advantage by making data-driven decisions that optimize profitability.**

- Innovation and Creativity: Insights from budget analysis can encourage studios to explore innovative and creative approaches within budget constraints, leading to unique and potentially successful projects.

Five Why's Analysis

Question	Explanation
1. Why do movies with higher budgets tend to have higher gross earnings?	1. Higher budgets allow for better marketing and wider distribution.
2. Why do better marketing and wider distribution lead to higher gross earnings?	2. More people become aware of the movie and have access to it, increasing ticket sales.
3. Why does increased awareness and access boost ticket sales?	3. Effective marketing creates anticipation and interest, while wide distribution ensures availability.
4. Why do anticipation and availability drive ticket sales?	4. People are more likely to watch a movie they are excited about and can easily find in theaters.
5. Why does excitement and accessibility result in higher earnings?	5. Positive buzz and ease of access lead to more viewers, higher sales, and thus greater earnings.

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Conclusion

Performing budget analysis and understanding its insights are essential for:

- **Making informed investment and budgeting decisions.**
- **Optimizing resource allocation.**
- **Benchmarking movie performance.**
- **Identifying market trends and audience preferences.**
- **Gaining a competitive advantage through data-driven strategies.**

Impact of Car Features

Project Overview

The car feature data analysis project aims to investigate the impact of various car features on price and profitability using a dataset with detailed information on car specifications, prices, and other relevant attributes. The project involves creating interactive dashboards and conducting in-depth analyses to derive actionable insights.

Objectives

1. **Feature Impact Analysis: Assess how different car features (e.g., horsepower, fuel efficiency, transmission type) affect car prices and profitability.**
2. **Visualizations: Create interactive visualizations to illustrate relationships between features and prices, including scatter plots, column charts, and pie charts.**
3. **Dashboard Development: Develop interactive dashboards to provide a comprehensive view of car price distributions, feature importance, and brand comparisons.**

Outliers Could Represent High-Value Insights

Real-World Scenarios and Exceptions Exist

Might Lead to the loss of valuable information

Results

- **Feature Relationships: Analyzed the impact of features such as engine power, fuel efficiency, and transmission type on car prices.**
- **Data Visualizations: Generated a variety of visualizations, including scatter plots and column charts, to highlight key relationships and trends in the data.**
- **Interactive Dashboards: Created dashboards that offer interactive insights into car features, pricing, and brand comparisons.**

3. Importance of Car Features in Determining Price

- The size and sign of each coefficient tell you how much the feature influences the car's price
- Positive coefficients ⟹ as the feature increases, so does the price
- Negative coefficients ⟹ as the feature increases, the price decreases

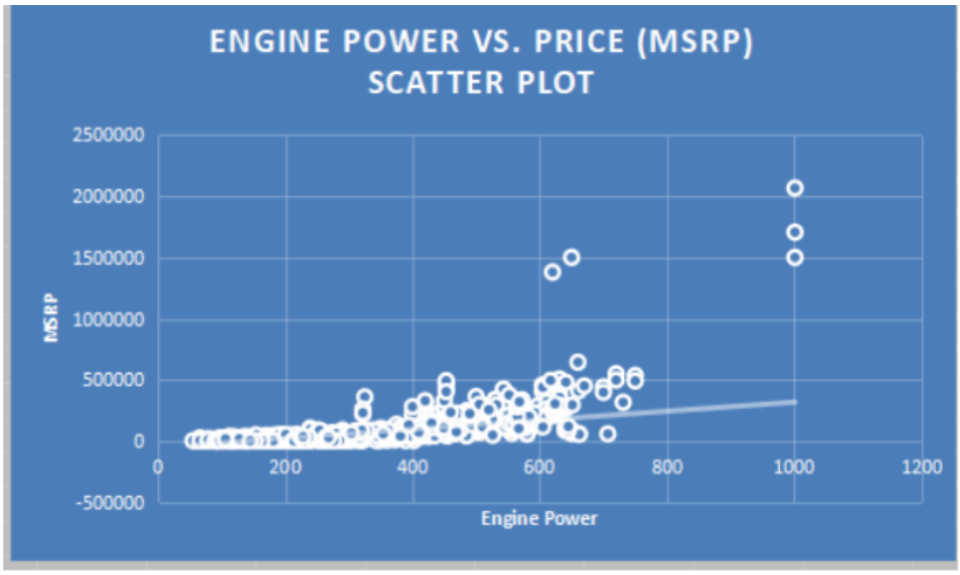
Feature	Coefficient
Engine HP	0.65837825
Engine Cylinders	0.540248512
Number of Doors	-0.144871758
highway MPG	-0.171939407
city mpg	-0.167552711
Popularity	-0.04876889

Insights:

- Horsepower has the highest positive coefficient, meaning that cars with higher horsepower tend to be more expensive
- MPG has a negative coefficient, it suggests that higher fuel efficiency is associated with lower prices, possibly due to smaller engines or more economical car models

Insights

- **Price Determinants: Engine power, fuel efficiency, and transmission type are significant factors influencing car prices.**
- **Brand and Body Style Impact: Different car brands and body styles show varied price distributions, affecting overall pricing strategies.**
- **Trend Analysis: Interactive dashboards reveal trends and correlations that help understand market dynamics and feature importance.**



Conclusion

The car feature data analysis project effectively provided insights into how car features influence pricing and profitability. The interactive dashboards and visualizations offer valuable tools for understanding market trends and making informed decisions regarding car features and pricing strategies.

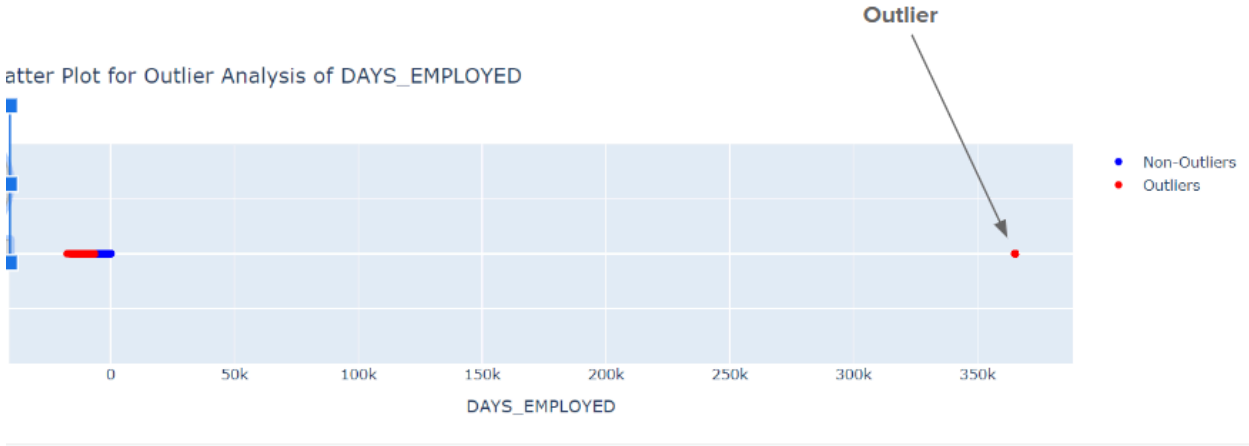
Bank Loan Case Study

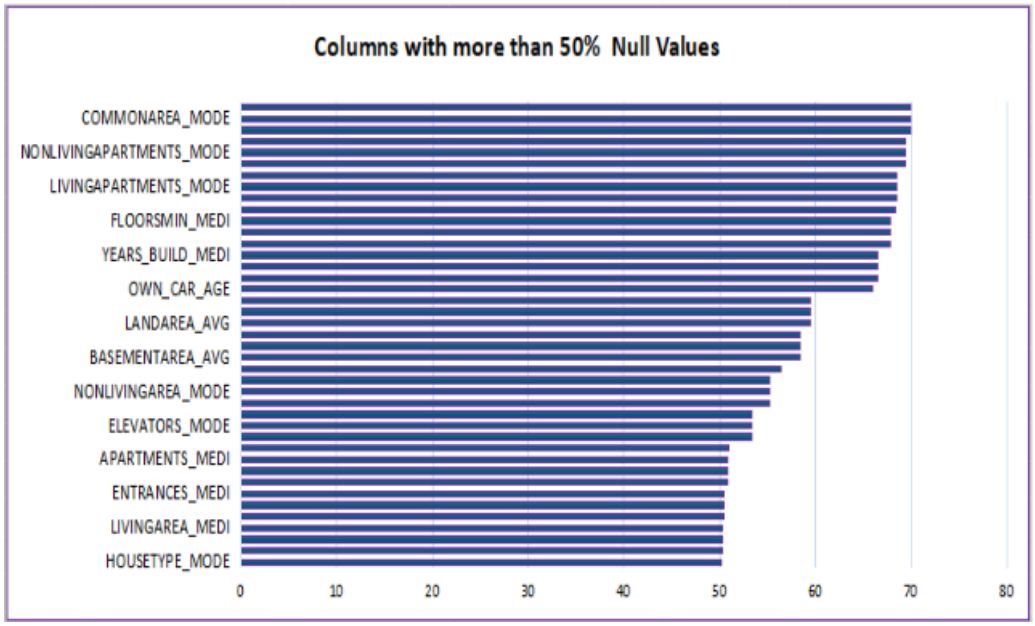
Project Overview

The bank loan case study involves analyzing loan application data to understand patterns, trends, and factors influencing loan approval and performance. The project utilizes various datasets to assess the impact of features on loan outcomes and aims to improve decision-making processes for loan applications.

Objectives

1. **Data Analysis: Analyze current and previous loan application data to identify trends, approval rates, and factors affecting loan outcomes.**
2. **Feature Impact: Evaluate how different features (e.g., income, credit score, loan amount) influence loan approval and performance.**
3. **Dashboard Creation: Develop interactive dashboards to visualize key metrics and insights from the data.**





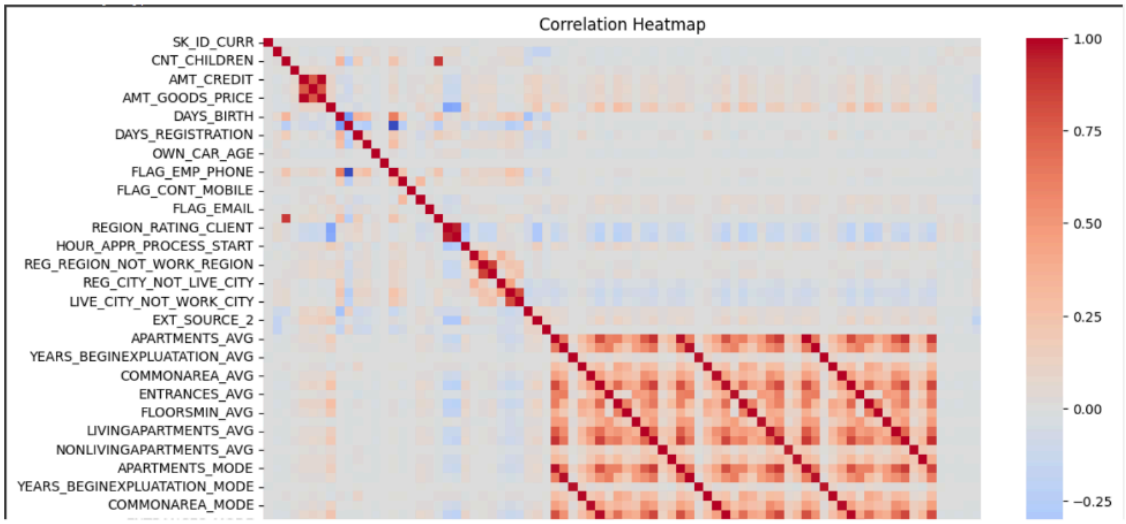
Results

- **Trend Identification: Identified significant trends in loan application data, including approval rates, common loan amounts, and applicant profiles.**
- **Feature Analysis: Determined the impact of various features on loan approval decisions and performance metrics.**
- **Visualization: Created interactive dashboards that provide clear and actionable insights into loan application patterns and outcomes.**

Insights

- **Approval Patterns: Certain features, such as credit score and income, have a strong correlation with loan approval rates.**
- **Application Trends: Trends in loan amounts and applicant profiles reveal important patterns that can inform future loan decisions.**
- **Dashboard Utility: Interactive dashboards effectively highlight key metrics and trends, aiding in better decision-making and strategy formulation.**

5. Identify Top Correlations for Different Scenarios



Conclusion

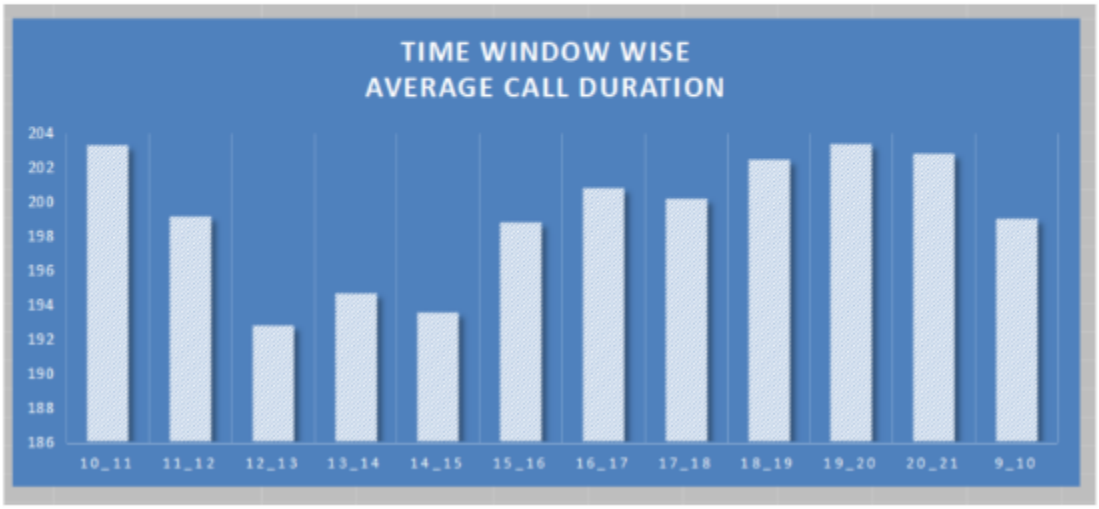
The case study successfully analyzed loan application data to uncover meaningful insights into approval patterns and influential features. The developed dashboards provide a valuable tool for visualizing and understanding loan trends, which can enhance the decision-making process and improve loan management strategies.

ABC Call Volume Trend Analysis

Project Overview

The project, titled "ABC Call Volume Trend Analysis," focuses on analyzing inbound call data to enhance customer experience and optimize manpower planning. The goal is to examine call trends, duration, and volume to propose effective manpower strategies, including adjustments for night shifts.

Average Call Duration	**198.6228**	3.310379574 mins



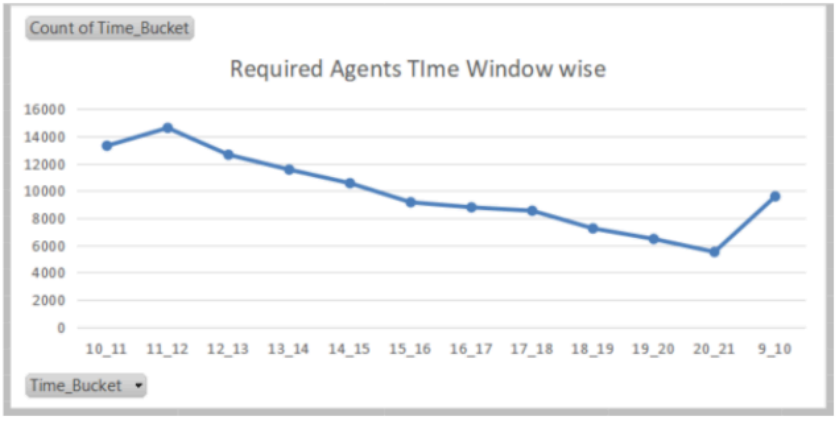
Objectives

1. **Analyze Call Volume and Duration: Examine trends in call volume and duration to understand peak times and average call duration.**
2. **Manpower Planning: Develop a plan to allocate manpower effectively to reduce the abandon rate from 30% to 10%.**
3. **Night Shift Planning: Propose a manpower plan for night shifts, aiming to maintain an abandon rate of 10% or lower.**

Results

- **Call Volume and Duration: Identified peak calling times and average call durations.**
- **Manpower Planning: Proposed a manpower allocation strategy to ensure adequate coverage during peak times.**
- **Night Shift Planning: Developed a staffing plan for night shifts to ensure service quality and meet the abandonment rate target.**

Time_Bucket	Required_Agents
10_11	4
11_12	5
12_13	4
13_14	4
14_15	4
15_16	3
16_17	3
17_18	3
18_19	2
19_20	2
20_21	2
9_10	3
Grand Total	**39**



Insights

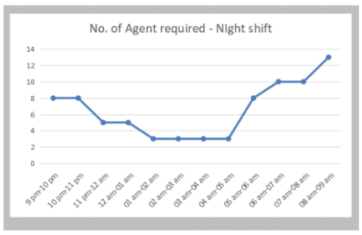
- **Peak Times: Specific times of day experience higher call volumes, necessitating increased manpower.**
- **Call Duration Trends: Call durations vary by time of day, impacting the required staffing levels.**
- **Abandonment Rate Management: Effective manpower planning can significantly reduce call abandonment rates.**

Nightshift Manpower Planning

04

- Proposed here is a manpower plan for each time bucket throughout the day, keeping the maximum abandon rate at 10%

Time - Bucket	Distribution of 30 calls	Percentage Distribution	No. of Calls	No. of Agent required
9 pm-10 pm	3	10%	137	8
10 pm-11 pm	3	10%	137	8
11 pm-12 am	2	7%	91	5
12 am-01 am	2	7%	91	5
01 am-02 am	1	3%	46	3
02 am-03 am	1	3%	46	3
03 am-04 am	1	3%	46	3
04 am-05 am	1	3%	46	3
05 am-06 am	3	10%	137	8
06 am-07 am	4	13%	183	10
07 am-08 am	4	13%	183	10
08 am-09 am	5	17%	229	13
	30	100%	1372	76



Insights and Results

Average Call Duration per Time Bucket

The graph shows the average duration of calls in different time buckets, indicating the busiest hours when agents spend the most time with customers. Overall it is about 3 minutes

- Understanding peak durations can help identify when customers are more engaged, which may align with specific marketing campaigns or service offerings

Call Volume Analysis

Identifying peak call volumes helps in better scheduling and allocation of resources during high-demand periods. It is high on most weekdays

- High call volumes may correspond with specific promotions, events, or issues, giving insights into customer behavior and service impact

Manpower Planning to Reduce Abandon Rate

Proper manpower planning ensures that enough agents are available to handle incoming calls, reducing the abandon rate and improving customer satisfaction

- Efficient resource allocation during peak times ensures a competitive edge by providing superior customer service

Night Shift Manpower Planning

Allocating the right number of agents during night shifts ensures that customer queries are handled even outside regular business hours, improving overall service

- Night shift planning reflects a company's commitment to customer service, potentially leading to increased customer loyalty

Conclusion

The analysis provided actionable insights into call volume trends and staffing needs, leading to optimized manpower planning for both regular and night shifts. Implementing the proposed plans should enhance customer service efficiency and achieve targeted abandonment rates.

Learnings related to specific skills in Excel, SQL, Python, and other tools from the projects

Excel

1. **Data Analysis and Visualization:**
    - **Pivot Tables and Charts: Mastered the use of pivot tables for summarizing and analyzing data, and created various charts (e.g., stacked column charts, scatter plots) to visualize trends and relationships.**
    - **Slicers and Interactivity: Learned to use slicers to add interactivity to dashboards, allowing for dynamic filtering and exploration of data.**
2. **Formulas and Functions:**
    - **COUNTIF and Other Functions: Gained proficiency in using functions like `COUNTIF` for counting specific data points (e.g., answered calls, abandoned calls), which is crucial for summarizing and analyzing data.**
3. **Dashboard Creation:**
    - **Interactive Dashboards: Developed skills in creating interactive dashboards that consolidate multiple visualizations and metrics, enhancing data presentation and decision-making.**

SQL

1. **Data Querying and Manipulation:**
    - **Complex Queries: Gained experience in writing complex SQL queries to extract and manipulate data, including filtering, aggregating, and joining tables.**
    - **Data Analysis: Used SQL to perform data analysis tasks such as calculating averages, counts, and other metrics essential for understanding trends and patterns in datasets.**
2. **Database Management:**

- Schema Design: Improved understanding of database schema design, including creating and managing tables and understanding relationships between different data entities.

Python

1. Data Preprocessing:
    - Handling Missing Values: Developed skills in using Python for data cleaning tasks, including handling missing values and outlier detection, which are critical for preparing data for analysis.
    - Data Manipulation: Gained experience with pandas for manipulating data, including merging, filtering, and transforming data frames.
2. Data Analysis and Visualization:
    - Plotly and Seaborn: Acquired skills in using Plotly and Seaborn for creating interactive and static visualizations, respectively. This includes generating scatter plots, line charts, and other visualizations to analyze data trends and relationships.
3. Automation and Efficiency:
    - Script Automation: Learned to automate repetitive data analysis tasks using Python scripts, improving efficiency and accuracy in handling large datasets.

General Skills

1. Integration of Tools:
    - Combining Excel, SQL, and Python: Gained experience in integrating these tools for comprehensive data analysis, using SQL for data extraction, Python for advanced analysis and automation, and Excel for visualization and reporting.
2. Data Interpretation and Communication:

- **Effective Communication: Improved ability to interpret data findings and communicate insights clearly through visualizations and dashboards, enhancing the ability to make data-driven decisions.**

3. **Problem-Solving:**
    - **Analytical Thinking: Enhanced problem-solving skills by addressing complex data challenges and deriving actionable insights from diverse datasets.**

<u>Presentation Skills</u>

1. **Structuring Presentations:**
    - **Clear Structure: Learned to create presentations with a clear and logical structure, including sections like Introduction, Data Overview, Analysis, Insights, and Conclusion. This helps in effectively communicating the project objectives and findings.**
    - **Concise Messaging: Developed the ability to distill complex data and insights into concise, easy-to-understand messages, which is crucial for engaging and informing your audience.**
2. **Visual Communication:**
    - **Effective Use of Visuals: Gained experience in using visuals (charts, graphs, dashboards) to support and enhance the presentation of data. This includes choosing the right types of visualizations to clearly convey trends, relationships, and key findings.**
    - **Interactive Dashboards: Incorporated interactive dashboards into presentations to allow stakeholders to explore data dynamically and gain deeper insights.**
3. **Presentation Scripts:**
    - **Script Writing: Improved skills in writing presentation scripts that are straightforward and tailored to the**

audience, using simpler language and avoiding complex terminology.
    - Engagement: Developed the ability to engage the audience by presenting data insights in a compelling and relatable manner, ensuring that key points are highlighted and understood.
4. Time Management:
    - Time Allocation: Learned to manage presentation time effectively, ensuring that each section is covered within the allotted time and that the presentation is concise yet comprehensive.
    - Focus on Key Points: Enhanced the ability to focus on key points and avoid unnecessary details, making the presentation more impactful and easier to follow.
5. Feedback and Iteration:
    - Incorporating Feedback: Improved the skill of incorporating feedback into presentations to address questions and concerns, and refine the content based on audience input.
    - Iterative Improvement: Gained experience in iterating on presentation materials to enhance clarity, accuracy, and effectiveness based on practice sessions and feedback.
6. Engaging Delivery:
    - Confident Delivery: Developed confidence in delivering presentations, ensuring that the message is communicated effectively and that the audience is engaged.
    - Interactive Elements: Used interactive elements (e.g., live demonstrations of dashboards) to maintain audience interest and involvement.

Appendix

Drive link(s)-
https://drive.google.com/drive/folders/1S3vmGhzl8EvoGMOl3nlJnDbvvUdZf0nI?usp=sharing